From Spikes to Symbols — architecture summary

biology → local perception → stable symbols → hierarchical composition → imagination → reasoning → uncertainty → continuous learning → neuromorphic scale

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BUILT TODAY (measured)

- A from-scratch spiking network in pure NumPy — **no gradients, no labels, no global error signal**.
- Two layers over four center-crossing **3×3 line primitives** (row, column, two diagonals).
- **L1**: pixel encoders with paired inhibitory neurons; **L2**: an overcomplete pool of pattern integrators, each with one trainable synapse per pixel.
- E/I identity carried by **synapse sign**; leaky integrate-and-fire membranes with refractory pacing.
- Learning is **local and spike-triggered**: a signed potentiate/depress rule on the firing neuron's own afferents, gated by a structural maturity brake.
- Competition is a **shared inhibitory neuron** that broadcasts a reset; losing capacity is conservatively **redistributed** into inactive gates (recruitment).
- A live FastAPI + Three.js **dashboard** visualizes spikes, membrane charge, and receptive-field formation in real time.

THE ARCHITECTURE (designed to enable)

- **Local receptive fields** replace one global view: many small circuits turn nearby events into features.
- Repeated exposure makes one neuron a **specialist**; its spike becomes a compact, reusable **symbol**.
- Higher layers learn **combinations of symbols**, not raw pixels — reuse, combinatorial generalization, modularity.
- Features are **size-agnostic**: scale by repeating circuits across space, not by changing the learning rule.
- Stable symbols reactivate without sensory input → **imagination** and counterfactual simulation.
- Reasoning becomes an **inspectable chain** of owned symbols and learned transitions; language describes it, but is not it.
- Response strength and competitive margin yield explicit **uncertainty and novelty** signals.

EXPECTED PROPERTIES

- **Continuous learning:** inference and learning share one substrate — no network-wide backward pass, no global loss.
- **Sparse compute:** only active pathways update; learning cost scales with events, not dataset size.
- **Data role shifts** from one frozen snapshot toward ongoing experience; gaps become measurable coverage, not diffuse bias.
- **Interpretability:** a concept traces back through the specialists that compose it.
- **Neuromorphic fit:** event-driven, local memory + compute, learning on-device, quiet when the world is unchanged.
- Ideal for **always-on robotics, edge devices, and adaptive sensors** under limited power.

The core local loop: `charge → fire → local feedforward update → shared-inhibitor competitive reset → feedback inhibition → repeat.` Every quantity a rule uses belongs to the neuron or synapse enacting it.

Predicted vs. measured: composition, imagination, reasoning, and hardware efficiency are **architectural predictions** the design pursues. Speed, energy, and memory gains on neuromorphic silicon are **future validation**, not benchmarked numbers.

Build intelligence from local systems that learn continuously, form explicit symbols, compose possibilities, trace their reasoning, and know the boundaries of their knowledge.